AERO50001 Aerodynamics 2

Tutorial Sheet 3

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1. Consider a vortex of strength Γ, at a height h above a horizontal plane wall in a uniform stream U_∞.

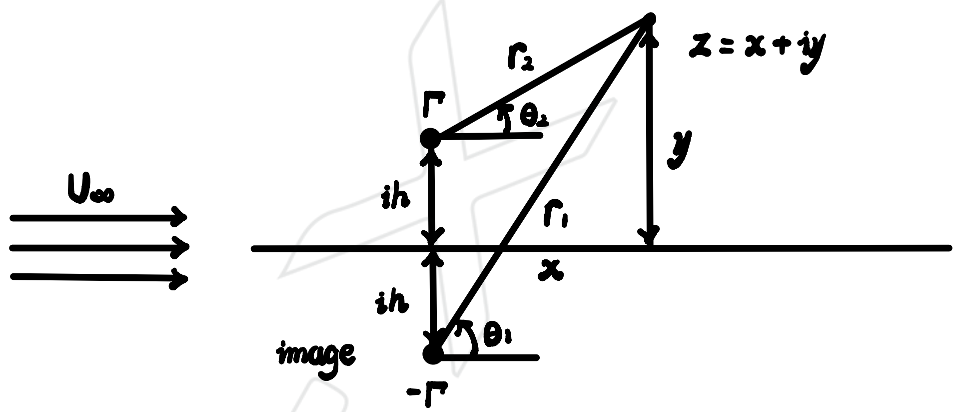
   (a) Show that the complex potential, $F(z)$, for the free-stream, the vortex and its image satisfies $\psi =$ constant on the wall.

   (b) Show that the flow speed on the wall is $u = U_\infty + \frac{\Gamma h}{\pi(x^2+h^2)}$.

   (c) Hence show that the pressure coefficient induced on the wall immediately beneath the vortex is $Cp = -\frac{2\Gamma}{\pi U_\infty h} - \frac{\Gamma^2}{(\pi U_\infty h)^2}$.

   [Note: $\ln\left(\frac{a-ib}{a+ib}\right) = -2i\tan^{-1}\left(\frac{b}{a}\right)$]

(a)



$$F(z) = U_\infty z - \frac{i\Gamma}{2\pi}\ln(z - ih) + \frac{i\Gamma}{2\pi}\ln(z-(-ih))$$

$$= U_\infty z + \frac{i\Gamma}{2\pi}\ln\left(\frac{z+ih}{z-ih}\right)$$

Let $z + ih = r_1 e^{i\theta_1}$, $z - ih = r_2 e^{i\theta_2}$

$$\therefore F(z) = U_\infty(x+iy) + \frac{i\Gamma}{2\pi}\ln\left(\frac{r_1}{r_2}e^{i(\theta_1-\theta_2)}\right)$$

$$= U_\infty x + iU_\infty y + \frac{i\Gamma}{2\pi}\ln\left(\frac{r_1}{r_2}\right) - \frac{\Gamma}{2\pi}(\theta_1 - \theta_2)$$

$\because F(z) = \phi + i\psi$

$\therefore \psi = U_\infty y + \frac{\Gamma}{2\pi}\ln(\frac{r_1}{r_2})$

$= U_\infty y + \frac{\Gamma}{2\pi}\ln(\frac{\sqrt{x^2+(y+h)^2}}{\sqrt{x^2+(y-h)^2}})$

At wall, $y=0$, $x \in \mathbb{R}$

$\therefore \psi = 0+0=0$

(b) $\phi = U_\infty x - \frac{\Gamma}{2\pi}(\theta_1 - \theta_2)$

$= U_\infty x - \frac{\Gamma}{2\pi}[\tan^{-1}(\frac{y+h}{x}) - \tan^{-1}(\frac{y-h}{x})]$

$\therefore u = \frac{\partial\phi}{\partial x}$

$= U_\infty + \frac{\Gamma}{2\pi}[\frac{y+h}{x^2}\frac{1}{1+(\frac{y+h}{x})^2} - \frac{y-h}{x^2}\frac{1}{1+(\frac{y-h}{x})^2}]$

$= U_\infty + \frac{\Gamma}{2\pi}[\frac{y+h}{x^2+(y+h)^2} - \frac{y-h}{x^2+(y-h)^2}]$

At wall, $y=0$, $x \in \mathbb{R}$

$\therefore u = U_\infty + \frac{\Gamma}{2\pi}[\frac{h}{x^2+h^2} - \frac{-h}{x^2+h^2}]$

$= U_\infty + \frac{\Gamma}{\pi}(\frac{h}{x^2+h^2})$

(c) $C_p = \dfrac{p - p_\infty}{\frac{1}{2}\rho U_\infty^2}$

Along the same streamline on the wall

$$p_0 = p + \frac{1}{2}\rho U^2 = p_\infty + \frac{1}{2}\rho U_\infty^2$$

$$\therefore p - p_\infty = \frac{1}{2}\rho U_\infty^2 - \frac{1}{2}\rho U^2$$

$$\therefore C_p = 1 - \left(\frac{U}{U_\infty}\right)^2$$

$$\therefore v = \frac{\partial\phi}{\partial y} = -\frac{\Gamma}{2\pi}\left[\frac{1}{x}\frac{1}{1+\left(\frac{y+h}{x}\right)^2} - \frac{1}{x}\frac{1}{1+\left(\frac{y-h}{x}\right)^2}\right]$$

$$= -\frac{\Gamma}{2\pi}\left[\frac{x}{x^2+(y+h)^2} - \frac{x}{x^2+(y-h)^2}\right]$$

$$\because u = U_\infty + \frac{\Gamma}{2\pi}\left[\frac{y+h}{x^2+(y+h)^2} - \frac{y-h}{x^2+(y-h)^2}\right]$$

Beneath the vortex, $x = 0$, $y = 0$

$$\therefore u = U_\infty + \frac{\Gamma}{2\pi}\left[\frac{1}{h} - \frac{1}{(-h)}\right] = U_\infty + \frac{\Gamma}{\pi}\left(\frac{1}{h}\right)$$

$$v = 0$$

$$\therefore U = \sqrt{u^2+v^2} = U_\infty + \frac{\Gamma}{\pi}\left(\frac{1}{h}\right)$$

$$\therefore C_p = 1 - \left(1 + \frac{\Gamma}{\pi U_\infty h}\right)^2 = 1 - \left(1 + \frac{2\Gamma}{\pi U_\infty h} + \frac{\Gamma^2}{(\pi U_\infty h)^2}\right)$$

$$= -\frac{2\Gamma}{\pi U_\infty h} - \frac{\Gamma^2}{(\pi U_\infty h)^2}$$

2. A flat plate aerofoil of zero thickness and 0.3 m chord is set at incidence to a flow of 50 m/s such that the airflow past it generates 500 N of lift per unit span.

(a) Calculate the angle of incidence required, assuming 2-D potential flow and sea-level conditions.

(b) Calculate the speed at the quarter-chord position on the upper surface of the aerofoil under these conditions.

(a) $C = 0.3m$, $U_\infty = 50m/s$, $L = 500N$, $\rho = 1.225 kg/m^3$

for Joukowski transformation.

$$\bar{z} = z + \frac{a^2}{z} = re^{i\theta} + \frac{a^2}{r}e^{-i\theta}$$

$$= r(\cos\theta + i\sin\theta) + \frac{a^2}{r}(\cos\theta - i\sin\theta)$$

$$= (r + \frac{a^2}{r})\cos\theta + i(r - \frac{a^2}{r})\sin\theta = \bar{x} + i\bar{y}$$

$\because \cos^2\theta + \sin^2\theta = 1$

$$\therefore \left[\frac{(r+\frac{a^2}{r})\cos\theta}{r+\frac{a^2}{r}}\right]^2 + \left[\frac{(r-\frac{a^2}{r})\sin\theta}{r-\frac{a^2}{r}}\right]^2 = 1$$

$$\therefore \left[\frac{\bar{x}}{r+\frac{a^2}{r}}\right]^2 + \left[\frac{\bar{y}}{r-\frac{a^2}{r}}\right]^2 = 1$$

$\therefore$ for flat plate with zero thickness and $r = R$

$a = R$

$\therefore C = 4R$

$$\because F(z) = U_\infty\left(ze^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z}\right) - \frac{i\Gamma}{2\pi}\ln z$$

Apply Joukowski transformation, $\bar{z} = z + \frac{a^2}{z}$

$$\therefore \frac{d\bar{z}}{dz} = 1 + \frac{a^2}{z^2}$$

$$\because u - iv = \frac{dF}{dz} = U_\infty\left(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}\right) - \frac{i\Gamma}{2\pi z}$$

$$\therefore \bar{u} - i\bar{v} = \frac{\frac{dF}{dz}}{\frac{d\bar{z}}{dz}} = \frac{U_\infty\left(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}\right) - \frac{i\Gamma}{2\pi z}}{1 - \frac{a^2}{z^2}}$$

At stagnation point, $\bar{u} - i\bar{v} = 0$

$$\therefore U_\infty\left(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}\right) - \frac{i\Gamma}{2\pi z} = 0$$

$$\therefore i\Gamma = 2\pi U_\infty\left(ze^{-i\alpha} - \frac{R^2}{z}e^{i\alpha}\right)$$

for Kutta condition, $r = R$ and $\theta = 0$

$$z = re^{i\theta} = R$$

$$\therefore i\Gamma = 2\pi U_\infty R\left(e^{-i\alpha} - e^{i\alpha}\right)$$

$$= 2\pi U_\infty R\left[(\cos\alpha - i\sin\alpha) - (\cos\alpha + i\sin\alpha)\right]$$

$$= 2\pi U_\infty R\left[i(-2\sin\alpha)\right]$$

$$\therefore \Gamma = -4\pi U_\infty R\sin\alpha$$

$\because L = -\rho U_\infty \Gamma$

$\therefore L = -\rho U_\infty(-4\pi U_\infty R\sin\alpha) = \pi\rho U_\infty^2 c\sin\alpha$

$\therefore \alpha = \arcsin\left(\frac{L}{\pi\rho U_\infty^2 c}\right) = 9.976°$

(b) $\bar{u} - i\bar{v} = \dfrac{u - iv}{\frac{d\bar{z}}{dz}} = \dfrac{U_\infty(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) - \frac{i\Gamma}{2\pi z}}{1 - \frac{a^2}{z^2}}$

for Joukowski transformation. at quarter-chord position,

for flat plate $a = R$

$|\bar{x}| = 2R\cos\theta = \frac{c}{4} = R$ $\quad\quad \therefore \bar{x} = -2R\cos\theta = R$

$\therefore \theta = \frac{2}{3}\pi$

$\therefore z = Re^{i\frac{2}{3}\pi}$

$\therefore u - iv = U_\infty(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) - \frac{i\Gamma}{2\pi z}$

$= U_\infty(e^{-i\alpha} - e^{-2i\theta}e^{i\alpha}) - i\frac{\Gamma}{2\pi R}e^{-i\theta}$

$\because \Gamma = -4\pi U_\infty R\sin\alpha$

$\therefore u - iv = U_\infty[(e^{-i\alpha} - e^{-2i\theta}e^{i\alpha}) - 2i\sin\alpha e^{-i\theta}]$

$$\therefore u - iv = U_\infty e^{-i\theta}(e^{i\theta}e^{-i\alpha} - e^{-i\theta}e^{i\alpha}) - i2U_\infty e^{-i\theta}\sin\alpha$$

$$= U_\infty e^{-i\theta}(2i\sin(\theta-\alpha)) - i2U_\infty e^{-i\theta}\sin\alpha$$

$$= i2U_\infty e^{-i\theta}(\sin(\theta-\alpha) - \sin\alpha)$$

$$\boxed{\begin{aligned} &= i2U_\infty(\cos\theta - i\sin\theta)(\sin(\theta-\alpha)-\sin\alpha) \\ &= iU_\infty[\sin(-\alpha)+\sin(2\theta-\alpha)-\sin(\alpha-\theta)-\sin(\alpha+\theta)-i\cos(\alpha)+i\cos(2\theta-\alpha) \\ &\quad -i\cos(\alpha-\theta)+i\cos(\theta+\alpha)] \\ &= U_\infty[\cos(\alpha)-\cos(2\theta-\alpha)+\cos(\alpha-\theta)-\cos(\alpha+\theta)] \\ &\quad + iU_\infty[\sin(-\alpha)+\sin(2\theta-\alpha)-\sin(\alpha-\theta)-\sin(\alpha+\theta)] \\ &= U_\infty[\cos(\alpha)-\cos(2\theta-\alpha)+\cos(\alpha-\theta)-\cos(\alpha+\theta)] \\ &\quad + iU_\infty[\sin(\alpha)+\sin(2\theta-\alpha)-\sin(\alpha-\theta)-\sin(\alpha+\theta)] \end{aligned}}$$

$$\therefore \tilde{u} - i\tilde{v} = \frac{u - iv}{1 - \frac{a^2}{z^2}} \boxed{= \frac{u-iv}{1-(\cos\theta - i\sin\theta)} = \frac{u-iv}{(1-\cos\theta)+i\sin\theta}}$$

$$= \frac{i2U_\infty e^{-i\theta}(\sin(\theta-\alpha)-\sin\alpha)}{1-e^{-2i\theta}}$$

$$\boxed{= \frac{u[(1-\cos\theta)-i\sin\theta)]}{(1-\cos\theta)^2+\sin^2\theta} - \frac{iv[(1-\cos\theta)-i\sin\theta)]}{(1-\cos\theta)^2+\sin^2\theta}}$$

$$= \frac{i2U_\infty(\sin(\theta-\alpha)-\sin\alpha)}{e^{i\theta}-e^{-i\theta}}$$

$$= U_\infty(\cos\alpha + \frac{\sin\alpha(1-\cos\theta)}{\sin\theta})$$

$$\boxed{= 64.25 - 0i}$$

$$= U_\infty(\cos\alpha \pm \sin\alpha\sqrt{\frac{1-\cos\theta}{1+\cos\theta}})$$

$$\therefore U = \sqrt{u^2+(v)^2} = 64.25\,(m/s)$$

3. Find the position of the stagnation point on a flat plate in a uniform stream at an angle of incidence α, when the circulation is given by the Kutta condition. Express the distance from the leading edge to the stagnation point as a fraction of the chord length.

$$\bar{z} = z + \frac{a^2}{z} = Re^{i\theta} + \frac{a^2}{R}e^{-i\theta}$$

$$= R(\cos\theta + i\sin\theta) + \frac{a^2}{R}(\cos\theta - i\sin\theta)$$

$$= (R + \frac{a^2}{R})\cos\theta + i(R - \frac{a^2}{R})\sin\theta = \bar{x} + i\bar{y}$$

$$\therefore \left(\frac{\bar{x}}{R + \frac{a^2}{R}}\right)^2 + \left(\frac{\bar{y}}{R - \frac{a^2}{R}}\right)^2 = 1$$

for flat plat. $a = R$

$$F(z) = U_\infty (ze^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z}) - \frac{i\Gamma}{2\pi}\ln z$$

$$\frac{dF}{d\bar{z}} = \frac{\frac{dF}{dz}}{\frac{d\bar{z}}{dz}} = \frac{U_\infty (e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) - \frac{i\Gamma}{2\pi z}}{1 - \frac{a^2}{z^2}} = \bar{u} - i\bar{v}$$

At stagnation point. $\bar{u} - i\bar{v} = 0$

$$\therefore U_\infty (e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) - \frac{i\Gamma}{2\pi z} = 0$$

for Kutta condition. $\theta = 0$ and $r = R$

$$\therefore z = R$$

$$\therefore U_\infty(e^{-i\alpha} - e^{i\alpha}) - \frac{i\Gamma}{2\pi R} = 0$$

$$\therefore \frac{i\Gamma}{2\pi R} = U_\infty(\cos\alpha - i\sin\alpha - \cos\alpha - i\sin\alpha) = -2iU_\infty\sin\alpha$$

$$\therefore \Gamma = -4\pi R U_\infty \sin\alpha$$

$$\therefore \bar{u} - i\bar{v} = \frac{U_\infty(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) + 2i\frac{R}{z}U_\infty\sin\alpha}{1 - \frac{a^2}{z^2}}$$

When $\bar{u} - i\bar{v} = 0$

$$U_\infty(e^{-i\alpha} - \frac{R^2}{z^2}e^{i\alpha}) + 2i\frac{R}{z}U_\infty\sin\alpha = 0$$

At the surface of the sphere, $z = Re^{i\theta}$

$$\therefore U_\infty(e^{-i\alpha} - e^{-i2\theta}e^{i\alpha}) + 2iU_\infty e^{-i\theta}\sin\alpha = 0$$

$$\therefore U_\infty e^{-i\theta}(e^{i\theta}e^{-i\alpha} - e^{-i\theta}e^{i\alpha} + 2i\sin\alpha) = 0$$

$$\therefore U_\infty e^{-i\theta}(e^{i(\theta-\alpha)} - e^{i(\alpha-\theta)} + 2i\sin\alpha) = 0$$

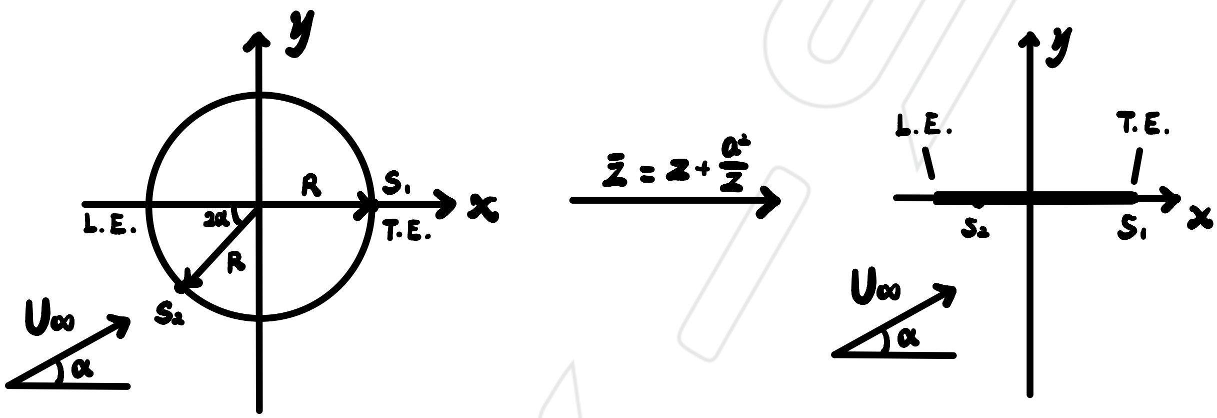
$$\therefore 2iU_\infty e^{-i\theta}(\sin(\theta-\alpha) + \sin\alpha) = 0$$

$$\because \sin(\theta-\alpha) + \sin\alpha = 0$$

$$\therefore \theta = 0 \text{ or } \pi + 2\alpha \text{ for } \theta \in [0, 2\pi]$$

for trailing edge at Kutta condition, $\theta = 0$, $z = R$

for leading edge, $\theta = \pi + 2\alpha$, $z = Re^{i\theta} = Re^{i(\pi+2\alpha)}$



$$\therefore \bar{z} = z + \frac{a^2}{z} = z + \frac{R^2}{z} = Re^{i\theta} + Re^{-i\theta} = R(e^{i\theta} + e^{-i\theta})$$

$$= R(2\cos\theta) = 2R\cos(\pi + 2\alpha) = -2R\cos(2\alpha)$$

When $\theta = 0$ and π, $z = R$ and $Re^{i\pi}$

$\therefore \bar{z} = \pm 2R \qquad \therefore \bar{c} = 2R - (-2R) = 4R$

$$\therefore \bar{d} = |-2R - \bar{z}| = 2R\,|\cos(2\alpha) - 1| = 2R[1 - \cos(2\alpha)]$$

$$\therefore \frac{\bar{d}}{\bar{c}} c = \frac{1}{2}[1 - \cos(2\alpha)]\, c$$

4. A cambered aerofoil, shown in the figure below, has zero thickness and is defined by

$$y = \frac{\epsilon x(c-x)}{c^2} \qquad \text{for} \quad 0 < x < c$$

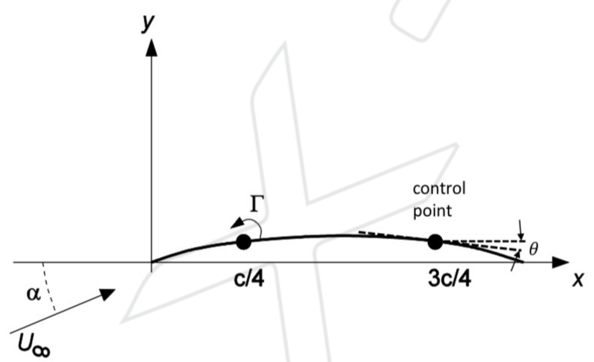
(a) Show that the camber to chord ratio $\lambda = \frac{\epsilon}{4c}$.

(b) The aerofoil is placed in a stream U_∞ at incidence α. By replacing the aerofoil by a single vortex of strength Γ at the quarter chord and satisfying the boundary condition of zero flow velocity normal to the surface at the three quarter chord point, show that

$$\Gamma = -\pi c U_\infty (\tan\theta\cos\alpha + \sin\alpha)$$

where θ is the angle between the surface tangent and the x-axis.

(c) Hence show that $C_L = 2\pi(2\lambda\cos\alpha + \sin\alpha)$.



(a) $\frac{dy}{dx} = \frac{\epsilon c - 2\epsilon x}{c^2}$

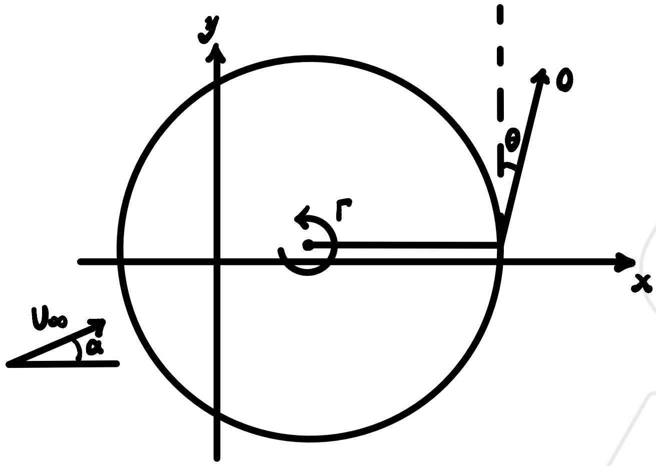
$\therefore$ when $\frac{dy}{dx} = 0$, y reach maximum, $x = \frac{c}{2}$

$\therefore \lambda = y|_{x=\frac{c}{2}} = \frac{\epsilon}{4c}$

(b) At the quarter chord, $x = \frac{1}{4}c$, $y = \frac{3}{16}\epsilon$

At the three quarter chord, $x = \frac{3}{4}c$, $y = \frac{3}{16}\epsilon$

$$\tan\theta = \left|\left.\frac{dy}{dx}\right|_{(x=\frac{3}{4}c,\ y=\frac{3}{16}\epsilon)}\right| = \left|-\frac{1}{2}\epsilon\right| = \frac{1}{2}\frac{\epsilon}{c}$$



$$F(z) = U_\infty z e^{-i\alpha} - \frac{i\Gamma}{2\pi}\ln z$$

$$= U_\infty r e^{i\theta} e^{-i\alpha} - \frac{i\Gamma}{2\pi}\ln r e^{i\theta}$$

$$= U_\infty r e^{i(\theta-\alpha)} - \frac{i\Gamma}{2\pi}\ln r + \frac{\Gamma}{2\pi}\theta$$

$$= U_\infty r\cos(\theta-\alpha) + iU_\infty r\sin(\theta-\alpha) - \frac{i\Gamma}{2\pi}\ln r + \frac{\Gamma}{2\pi}\theta$$

$$= \phi + i\psi$$

$$\therefore \phi = U_\infty r\cos(\theta-\alpha) + \frac{\Gamma}{2\pi}\theta$$

$$\psi = U_\infty r\sin(\theta-\alpha) - \frac{\Gamma}{2\pi}\ln r$$

$$\therefore V_r = \frac{\partial\phi}{\partial r} = U_\infty\cos(\theta-\alpha)$$

$$V_\theta = -\frac{\partial\psi}{\partial r} = -U_\infty\sin(\theta-\alpha) + \frac{\Gamma}{2\pi r}$$

At the three quarter chord. $r = \frac{3}{4}c - \frac{1}{4}c = \frac{1}{2}c$, $\theta = 0$

$\therefore V_r = u = U_\infty \cos(-\alpha) = U_\infty \cos\alpha$

$V_\theta = v = -U_\infty \sin(-\alpha) + \frac{\Gamma}{\pi c} = U_\infty \sin\alpha + \frac{\Gamma}{\pi c}$

$\because u\sin\theta + v\cos\theta = 0 \longleftarrow \vec{U}\cdot\vec{\eta} = (u, v)\cdot(\sin\theta, \cos\theta) = 0$

$\therefore u\tan\theta + v = 0$

$\therefore U_\infty \tan\theta\cos\alpha + U_\infty \sin\alpha + \frac{\Gamma}{\pi c} = 0$

$\therefore U_\infty(\tan\theta\cos\alpha + \sin\alpha) = -\frac{\Gamma}{\pi c}$

$\therefore \Gamma = -\pi c U_\infty(\tan\theta\cos\alpha + \sin\alpha)$

(c) $L = -\rho U_\infty \Gamma = \rho\pi c U_\infty^2(\tan\theta\cos\alpha + \sin\alpha)$

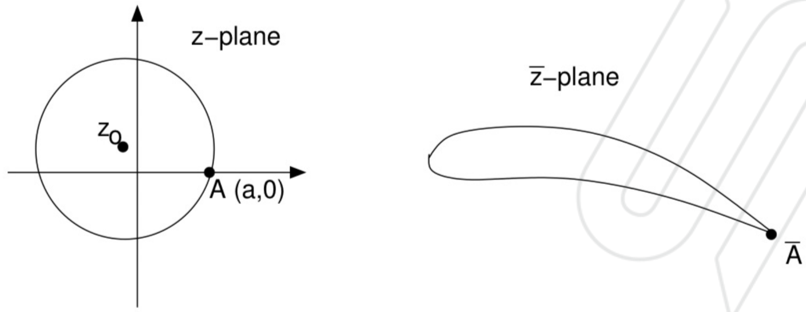
$\therefore C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 c} = 2\pi(\tan\theta\cos\alpha + \sin\alpha)$

$\because 2\lambda = \frac{\varepsilon}{2c} = \tan\theta$

$\therefore C_L = 2\pi(2\lambda\cos\alpha + \sin\alpha)$

5. (a) Write down the complex potential for the flow around a circular cylinder of radius R, centre z_0 and circulation Γ when the flow far from it has uniform speed U at an incidence angle α.

The transformation $\overline{z} = z + \frac{a^2}{z}$ maps the circular cylinder shown on the left below into the cambered aerofoil on the right. Point A on the circle maps onto the trailing edge of the aerofoil, $\overline{A}$.



(b) The Kutta condition requires the velocity at the trailing edge to be finite. Explain why this implies that $\frac{dF}{dz}$ (i.e. in the circle-plane) must be zero at point A.

(c) Hence obtain an expression for the circulation Γ around the aerofoil that satisfies the Kutta condition when the aerofoil is placed in oncoming flow with speed U at an incidence angle α.

(d) If $z_0 = -0.1a + 0.2ai$ (so that $(a - z_0) = 1.118a\, e^{i\gamma}$ where $\gamma = -0.18$ radians) and $R = 1.118a$, show that this becomes $\Gamma = -4\pi U a 1.118 \sin(\alpha - \gamma)$.

(e) Explain how this circulation is generated when the motion starts from rest in an inviscid fluid.

(a) $F(z) = U_\infty \left[(z - z_0) e^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z - z_0} \right] - \frac{i\Gamma}{2\pi} \ln(z - z_0)$

(b) otherwise we could not use inviscid theory to predict realistic flow due to the effect of flow seperation at the trailing edge. At $(a, 0)$, $\frac{d\bar{z}}{dz} = 1 - \frac{a^2}{z^2} = 0$ $\therefore$ only $u - iv = 0$, $\bar{u} - i\bar{v} = 0$ not ∞

(c) $\frac{dF}{dz} = U_\infty \left[e^{-i\alpha} - \frac{R^2}{(z - z_0)^2} e^{i\alpha} \right] - \frac{i\Gamma}{2\pi (z - z_0)}$

$= u - iv$

At $A(a,0)$, $z=a$, $u-iv=0$

$$\therefore U_\infty\left[e^{-i\alpha}-\frac{R^2}{(z-z_0)^2}e^{i\alpha}\right]-\frac{i\Gamma}{2\pi(z-z_0)}$$

$$=U_\infty\left[e^{-i\alpha}-\frac{R^2}{(a-z_0)^2}e^{i\alpha}\right]-\frac{i\Gamma}{2\pi(a-z_0)}=0$$

$$\therefore i\Gamma=2U_\infty\pi(a-z_0)\left[e^{-i\alpha}-\frac{R^2}{(a-z_0)^2}e^{i\alpha}\right]$$

$$\therefore \Gamma=-2iU_\infty\pi\left[(a-z_0)e^{-i\alpha}-\frac{R^2}{a-z_0}e^{i\alpha}\right]$$

(d) $$\Gamma=-2iU_\infty\pi\left(1.118ae^{i\gamma}e^{-i\alpha}-\frac{(1.118a)^2}{1.118a}e^{-i\gamma}e^{i\alpha}\right)$$

$$=-2iU_\infty\pi(1.118a)(e^{i\gamma}e^{-i\alpha}-e^{-i\gamma}e^{i\alpha})$$

$$=-2iU_\infty\pi(1.118a)$$

$$[(\cos\gamma+i\sin\gamma)(\cos\alpha-i\sin\alpha)-(\cos\gamma-i\sin\gamma)(\cos\alpha+i\sin\alpha)]$$

$$=-2iU_\infty\pi(1.118a)(2i\sin\gamma\cos\alpha-2i\cos\gamma\sin\alpha)$$

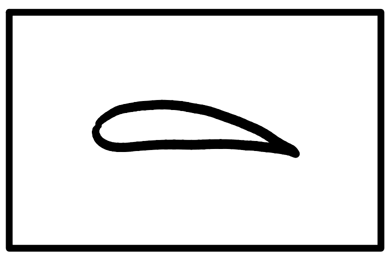
$$=-2iU_\infty\pi(1.118a)\,i\,[\sin(\gamma-\alpha)+\sin(\gamma+\alpha)-\sin(\alpha-\gamma)-\sin(\alpha+\gamma)]$$

$$=-2iU_\infty\pi(1.118a)\,i\,[-2\sin(\alpha-\gamma)]$$

$$=-4\pi U_\infty a\,1.118\sin(\alpha-\gamma)$$

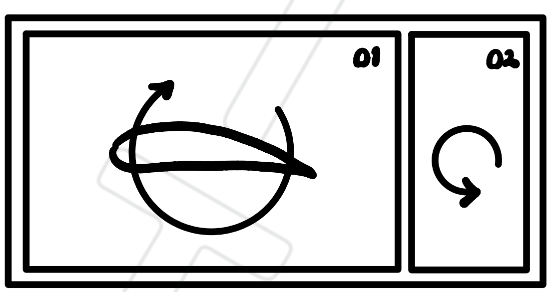
(e) According to the Kelvin's theorem, $\frac{d\Gamma}{dt} = 0$

① At rest :



$\Gamma_1 = 0$ at $t = 0$

② The flow decelerates from large speed around the T.E. to the stagnation point, results seperation occurs and an anticlockwise vortex is shed :



$\Gamma_2 = \Gamma_{01} + \Gamma_{02} = 0$ at $t = T$

$\therefore \Gamma_{01} = -\Gamma_{02}$

$\therefore$ Circulation is generated

Numerical answers:

1. - 2. (a) 10 degrees (b) 64.3 m/s

3. $\frac{1}{2}(1 - \cos 2\alpha) \times$ the chord length 4. -

5. (a) $F(z) = U_\infty \left((z - z_0)e^{-i\alpha} + \frac{R^2 e^{i\alpha}}{z - z_0} \right) - \frac{i\Gamma}{2\pi} \ln(z - z_0)$ (c) $\Gamma = -2\pi i U \left((a - z_0)e^{-i\alpha} - \frac{R^2 e^{i\alpha}}{a - z_0} \right)$